

29. Morphological Operation using Erosion

AIM

To perform morphological erosion on an image using OpenCV.

PROCEDURE

1. Read the image.
2. Convert it to grayscale.
3. Create a morphological kernel.
4. Apply cv2.erode().
5. Display the eroded image.

PROGRAM

```
import cv2
import numpy as np

# Read image
img = cv2.imread(r"C:\Users\chait\OneDrive\Documents\Desktop\cv lab image.jpg")

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply erosion
eroded = cv2.erode(gray, kernel, iterations=1)

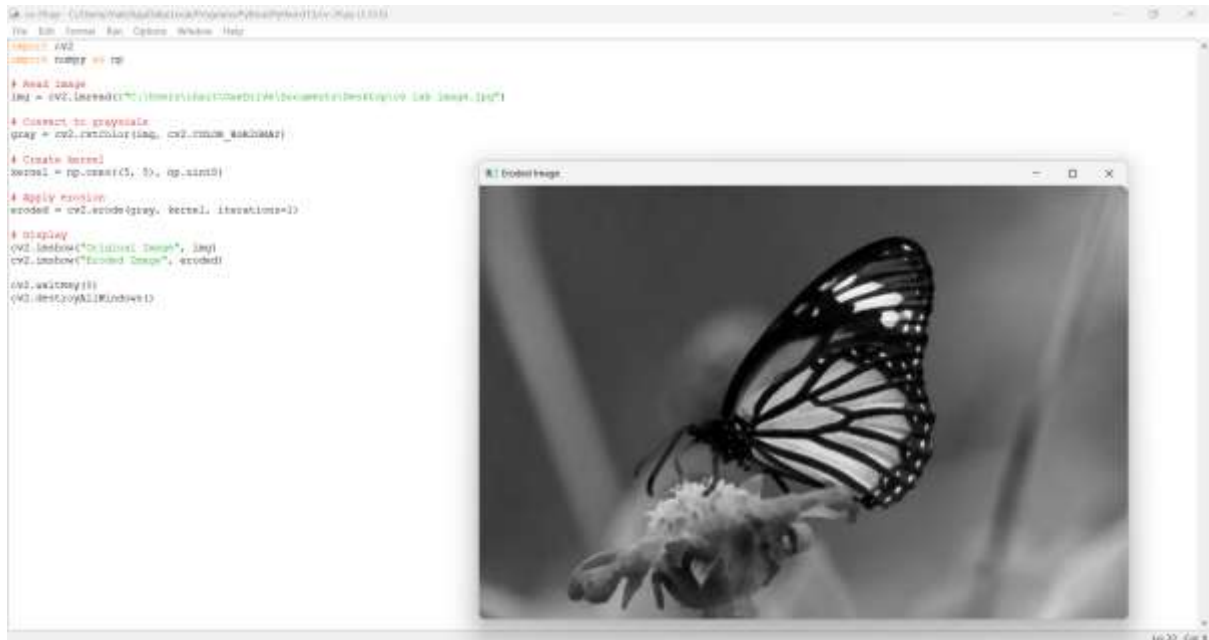
# Display
cv2.imshow("Original Image", img)
cv2.imshow("Eroded Image", eroded)
```

```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

OUTPUT

The objects in the image become thinner after erosion.



Explanation

Erosion removes pixels around the boundaries of objects. It makes bright objects smaller and is useful for removing small noise and separating connected objects.